The Journal of Open Source Software

spotgp: Physically Interpretable Gaussian Process Kernels for Starspot Variability

Jessica Birky [1,2¶], **Eric Agol** [1], **Rory Barnes** [1], **James R. A. Davenport** [1], **and Sean Matt** [2]

1 Department of Astronomy, University of Washington, Seattle, WA, USA **2** Homer L. Dodge Department of Physics and Astronomy, University of Oklahoma, Norman, OK 73019, USA ¶ Corresponding author

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor: Open Journals

Reviewers:

- @openjournals

Submitted: 01 January 1970
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

Summary

Large-scale time-domain surveys, such as *Kepler*, *K2*, *TESS* and *PLATO*, are enabling astronomers to study the dynamic surfaces of millions of stars. In particular, low-mass stars are covered in dark magnetic spots that rotate in and out of view, as well as emerge, grow, and decay stochastically over time. These spots cause a star's observed brightness to vary quasi-periodically with time, encoding not only how fast a star spins but also how it is tilted relative to our line of sight, whether its equator rotates faster than its poles (differential rotation), and how long its spots live.

`spotgp` is a Python package for modeling spotted lightcurves and statistically inferring spot properties from stellar lightcurves using Gaussian processes. A Gaussian process (GP) (Rasmussen & Williams, 2006) is a probabilistic model whose covariance function, or kernel, specifies how strongly the brightness at two different times is correlated. The `spotgp` kernel is derived analytically from a model of circular spots on a differentially rotating star (Kipping, 2012) (Figure 1), so each of its hyperparameters is a physical quantity: the equatorial rotation period $P_{\rm eq}$, the differential rotation shear κ, the stellar inclination I, the spot lifetime $\ell_{\rm spot}$, the spot emergence and decay timescales, and an amplitude σ_k (Birky et al., 2026). Fitting the kernel to a lightcurve therefore gives posterior distributions over stellar and spot properties directly.

The package is built on `JAX` (Bradbury et al., 2018) for just-in-time compilation, automatic differentiation, and GPU execution, and includes a forward light-curve simulator, a scalable GP solver, and interfaces to gradient-based and nested Markov chain Monte Carlo methods for posterior sampling.

DRAFT

The Journal of Open Source Software

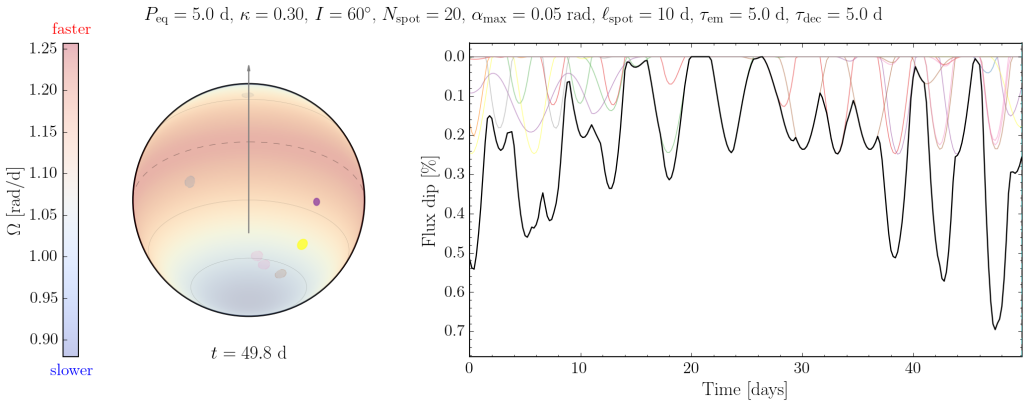


Figure 1: The forward model underlying the `spotgp` kernel. Circular spots emerge and decay at different latitudes on a differentially rotating, inclined star (left; color shows the local rotation rate), and each contributes a flux deficit as it rotates across the visible disk (right; thin lines are individual spots, the black line is the total). The GP kernel is the analytic autocovariance of this process.

Statement of need

Rotation periods are now measured routinely for large samples of stars, but stellar inclination, differential rotation, and spot lifetimes are not, despite their importance for stellar dynamo theory, gyrochronology, and the treatment of stellar activity as noise in exoplanet surveys. These quantities have mostly been measured with spectroscopy, asteroseismology, or spot-crossing transits, which restricts them to stars with high-resolution spectra, detectable oscillations, or transiting planets. Measuring them from photometry alone has proven difficult. A blind hare-and-hounds exercise found little correlation between the reported and true differential rotation for existing photometric methods (Aigrain et al., 2015), an information-content analysis of analytic spot models suggests that differential rotation is nearly inaccessible from lightcurves alone (Basri & Shah, 2020), and the hyperparameters of the widely used quasi-periodic GP kernel do not map cleanly onto spot properties (Nicholson & Aigrain, 2022). `spotgp` is designed to make these quantities measurable from a single photometric time series, and is aimed at researchers analyzing stellar lightcurves who want physically interpretable spot and rotation parameters with probabilistic uncertainties.

State of the field

There are a number of tools available for forward modeling lightcurves of spotted stars (e.g., `macula` (Kipping, 2012), `butterpy` (Claytor et al., 2022), `starry` (Luger et al., 2019)) using analytic models of the flux blocked by spots on a rotating star. Fitting them to data means sampling the properties of every individual spot, and they provide no covariance function for GP inference.

Simultaneously, there are also a number of packages specialized for fitting GPs to astronomical time series. `celerite` and `celerite2` (Foreman-Mackey et al., 2017; Foreman-Mackey, 2018), `george` (Ambikasaran et al., 2016), `tinygp` (Foreman-Mackey, 2024), and `exoplanet` (Foreman-Mackey et al., 2021) provide fast solvers together with quasi-periodic or damped-oscillator kernels that fit spotted-star lightcurves well and recover rotation periods (Angus et al., 2018).

However, the kernel models implemented in these packages are not designed specifically for describing the physical properties of starspots. Apart from the period, their hyperparameters are not tied to spot physics, and none includes differential rotation. The closest relative is `starry_process` (Luger et al., 2021), an interpretable GP whose hyperparameters describe

DRAFT

the latitude, size, and contrast distribution of spots on a static surface; it does not model spot emergence and decay or differential rotation, which are the temporal effects `spotgp` targets.

Software design

`spotgp` is designed to be modular and flexible, allowing users to easily customize their model for different physical assumptions, while also being optimized for computational speed. Figure Figure 2 shows an overview of the architecture: a model layer (yellow) where the physical parameters (purple) and assumptions of the model are defined, a simulation layer (orange) containing the framework to generate sythetic light curves, a kernel layer (blue) containing the tools to compute the GP kernel matrix, a data layer (green) for processing the lightcurve data, and finally, an inference layer implementing the GP solver and interface with the MCMC samplers.

The kernel function describes the autocorrelation of the flux as a function of time lag τ, which factorizes as

$$k(\tau) = \sigma_k^2 \int \underbrace{R_\Gamma(\tau)}_{\substack{\text{Envelope}\\ \text{function}}} \underbrace{\left[|c_0(I,\Phi)|^2 + 2\sum_{n\geq 1} |c_n(I,\Phi)|^2 \cos\left(n\,\omega_0(\Phi)\,\tau\right) \right]}_{\substack{\text{Visibility}\\ \text{function}}} \underbrace{p(\Phi)\,d\Phi}_{\substack{\text{Latitude}\\ \text{distribution}}},$$

where R_Γ is the autocorrelation of the spot emergence-and-decay envelope, c_n are Fourier coefficients of the spot visibility function, $\omega_0(\Phi) = 2\pi(1 - \kappa \sin^2 \Phi)/P_{\text{eq}}$ is the latitude-dependent rotation frequency, and $p(\Phi)$ is the spot latitude distribution.

Each factor is a base class (`EnvelopeFunction`, `VisibilityFunction`, `LatitudeDistributionFunction`) which defines the spot evolution model (`SpotEvolutionModel`). This modularity allows for users to create custom inheritance clases allowing the user to change one physical assumption without touching the kernel evaluation, the latitude quadrature, the compiled likelihood, or the samplers.

The same `SpotEvolutionModel` can be used for the time series forward model (`LightcurveModel`) to generate synthetic lightcurves or can be used directly to compute the GP covariance matrix from the kernel function. Because the simulation layer and kernel layer share the same `SpotEvolutionModel`, it is easy to validate the analytic kernel against simulations with identical assumptions (through the `NumericalKernel` class), allowing the user to do quick sanity checks when implementing new analytic equations into their model.

DRAFT

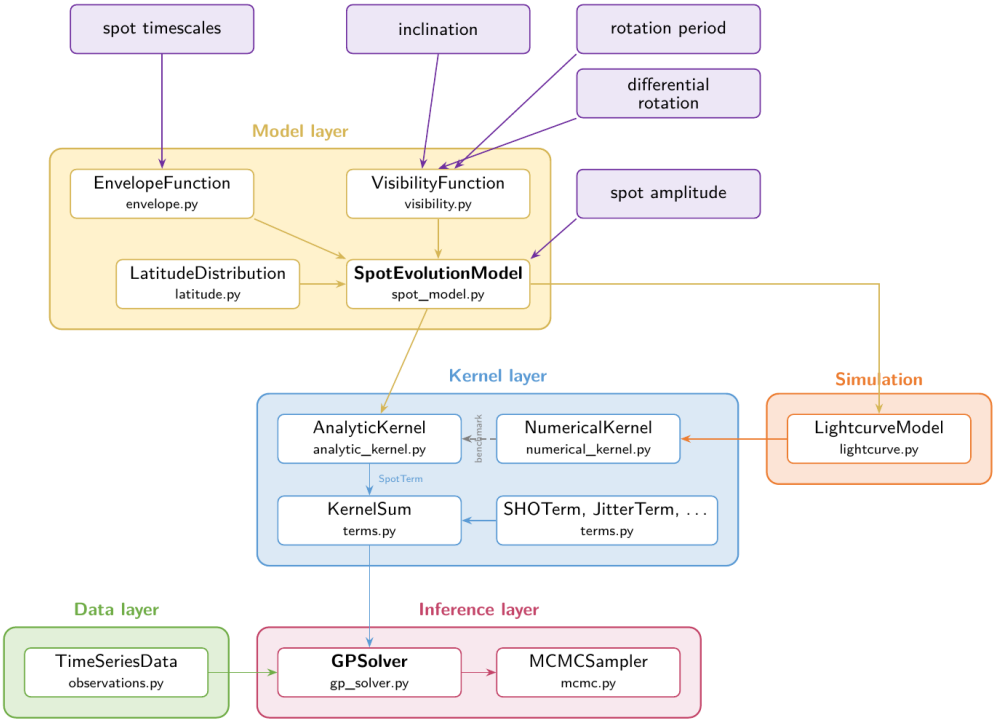


Figure 2: Architecture of the `spotgp` package.

All modules are implemented `JAX` so that the whole pipeline (latitude quadrature, harmonic sum, covariance assembly, Cholesky factorization, and log-likelihood) is one JIT-compiled, differentiable function. It can be sampled with the No-U-Turn Sampler (Hoffman & Gelman, 2014) through `BlackJAX` (Cabezas et al., 2024) or with nested sampling through `dynesty` (Speagle, 2020).

Naive GP solvers perform poorly with the number of datapoints N, scaling as $\mathcal{O}(N^3)$ (Rasmussen & Williams, 2006). However, for finite-lived spots our covariance matrix becomes banded with a bandwidth b set by the spot lifetimes and cadence. Our package exploits this structure to save memory and computation time implement a blocked banded Cholesky factorization (Golub & Van Loan, 2013) written with `jax.lax.scan` which runs in $\mathcal{O}(Nb^2)$ time.

Research impact statement

`spotgp` is the computational basis of a companion paper (Birky et al., 2026) that derives the kernel, verifies the derivation with `SymPy`, validates the kernel against Monte Carlo simulations, and uses MCMC fits and Cramér–Rao bounds on simulated lightcurves to map how well rotation, inclination, and spot parameters can be recovered as a function of cadence, baseline, and noise level. That paper is fully reproducible: every figure is generated by `spotgp` scripts under `showyourwork`, including benchmarks of the `spotgp` solver against `tinygp` and `george`.

`spotgp` has been developed on GitHub since July 2023 with comprehensive unit tests, run in continuous integration on Linux and macOS for Python 3.10–3.12. It is available through PyPI and can e installed via `pip install spotgp`.

DRAFT

AI usage disclosure

Generative AI tools (Anthropic's Claude, via the Claude Code assistant) were used as a coding and writing aid during the development of `spotgp` and the preparation of this manuscript: to draft and edit prose in the documentation and in this paper, to scaffold unit tests and boilerplate, and to assist with refactoring. All scientific claims, derivations, and results were produced and checked by the authors.

Acknowledgements

E.A. was supported by the National Aeronautics and Space Administration Exoplanet Research Program (NASA XRP) grant 80NSSC21K1111. This work was performed in part by the Virtual Planetary Laboratory Team, which is a member of the NASA Nexus for Exoplanet System Science, and funded via the NASA Astrobiology Program ICAR Grant 80NSSC23K1398.

References

Aigrain, S., Llama, J., Ceillier, T., Chagas, M. L. das, Davenport, J. R. A., García, R. A., Hay, K. L., Lanza, A. F., McQuillan, A., Mazeh, T., Medeiros, J. R. de, Nielsen, M. B., & Reinhold, T. (2015). Testing the recovery of stellar rotation signals from Kepler light curves using a blind hare-and-hounds exercise. *Monthly Notices of the Royal Astronomical Society*, *450*(3), 3211–3226. https://doi.org/10.1093/mnras/stv853

Ambikasaran, S., Foreman-Mackey, D., Greengard, L., Hogg, D. W., & O'Neil, M. (2016). Fast direct methods for Gaussian processes. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *38*(2), 252–265. https://doi.org/10.1109/TPAMI.2015.2448083

Angus, R., Morton, T., Aigrain, S., Foreman-Mackey, D., & Rajpaul, V. (2018). Inferring probabilistic stellar rotation periods using Gaussian processes. *Monthly Notices of the Royal Astronomical Society*, *474*(2), 2094–2108. https://doi.org/10.1093/mnras/stx2109

Basri, G., & Shah, R. (2020). *The information content in analytic spot models of broadband precision light curves. II. Spot distributions and lifetimes, global and differential rotation*. arXiv. https://doi.org/10.48550/arXiv.2008.04969

Birky, J., Agol, E., Barnes, R., & Davenport, J. R. A. (2026). *Physically motivated Gaussian processes for stellar variability I: A starspot kernel with differential rotation*. Manuscript in preparation.

Bradbury, J., Frostig, R., Hawkins, P., Johnson, M. J., Leary, C., Maclaurin, D., Necula, G., Paszke, A., VanderPlas, J., Wanderman-Milne, S., & Zhang, Q. (2018). *JAX: Composable transformations of Python+NumPy programs*. http://github.com/jax-ml/jax

Cabezas, A., Corenflos, A., Lao, J., Louf, R., Carnec, A., Chaudhari, K., Cohn-Gordon, R., Coullon, J., Deng, W., Duffield, S., Durán-Martín, G., Elantkowski, M., Foreman-Mackey, D., Gregori, M., Iguaran, C., Kumar, R., Lysy, M., Murphy, K., Orduz, J. C., … Zinkov, R. (2024). *BlackJAX: Composable Bayesian inference in JAX*. arXiv. https://doi.org/10.48550/arXiv.2402.10797

Claytor, Z. R., Saders, J. L. van, Llama, J., Sadowski, P., Quach, B., & Avallone, E. A. (2022). Recovery of TESS stellar rotation periods using deep learning. *The Astrophysical Journal*, *927*(2), 219. https://doi.org/10.3847/1538-4357/ac498f

Foreman-Mackey, D. (2018). Scalable backpropagation for Gaussian processes using celerite. *Research Notes of the American Astronomical Society*, *2*(1), 31. https://doi.org/10.3847/2515-5172/aaaf6c

Foreman-Mackey, D. (2024). *Tinygp: The tiniest of Gaussian process libraries*. https:

DRAFT

//github.com/dfm/tinygp

Foreman-Mackey, D., Agol, E., Ambikasaran, S., & Angus, R. (2017). Fast and scalable Gaussian process modeling with applications to astronomical time series. *The Astronomical Journal*, *154*(6), 220. https://doi.org/10.3847/1538-3881/aa9332

Foreman-Mackey, D., Luger, R., Agol, E., Barclay, T., Bouma, L., Brandt, T., Czekala, I., David, T., Dong, J., Gilbert, E., Gordon, T., Hedges, C., Hey, D., Morris, B., Price-Whelan, A., & Savel, A. (2021). Exoplanet: Gradient-based probabilistic inference for exoplanet data & other astronomical time series. *The Journal of Open Source Software*, *6*(62), 3285. https://doi.org/10.21105/joss.03285

Golub, G. H., & Van Loan, C. F. (2013). *Matrix computations* (4th ed.). Johns Hopkins University Press. ISBN: 9781421407944

Hoffman, M. D., & Gelman, A. (2014). The no-U-turn sampler: Adaptively setting path lengths in Hamiltonian Monte Carlo. *Journal of Machine Learning Research*, *15*(47), 1593–1623. http://jmlr.org/papers/v15/hoffman14a.html

Kipping, D. M. (2012). An analytic model for rotational modulations in the photometry of spotted stars. *Monthly Notices of the Royal Astronomical Society*, *427*(3), 2487–2511. https://doi.org/10.1111/j.1365-2966.2012.22124.x

Luger, R., Agol, E., Foreman-Mackey, D., Fleming, D. P., Lustig-Yaeger, J., & Deitrick, R. (2019). Starry: Analytic occultation light curves. *The Astronomical Journal*, *157*(2), 64. https://doi.org/10.3847/1538-3881/aae8e5

Luger, R., Foreman-Mackey, D., & Hedges, C. (2021). Mapping stellar surfaces. II. An interpretable Gaussian process model for light curves. *The Astronomical Journal*, *162*(3), 124. https://doi.org/10.3847/1538-3881/abfdb9

Nicholson, B. A., & Aigrain, S. (2022). Quasi-periodic Gaussian processes for stellar activity: From physical to kernel parameters. *Monthly Notices of the Royal Astronomical Society*, *515*(4), 5251–5266. https://doi.org/10.1093/mnras/stac2097

Rasmussen, C. E., & Williams, C. K. I. (2006). *Gaussian processes for machine learning*. MIT Press. https://doi.org/10.7551/mitpress/3206.001.0001

Speagle, J. S. (2020). Dynesty: A dynamic nested sampling package for estimating Bayesian posteriors and evidences. *Monthly Notices of the Royal Astronomical Society*, *493*(3), 3132–3158. https://doi.org/10.1093/mnras/staa278

DRAFT